

Basic Geometry

In this article we will consider basic operations on points in Euclidean space which maintains the foundation of the whole analytical geometry. We will consider for each point $\mathbf{r}$ the vector $\vec{\mathbf{r}}$ directed from $\mathbf{0}$ to $\mathbf{r}$. Later we will not distinguish between $\mathbf{r}$ and $\vec{\mathbf{r}}$ and use the term **point** as a synonym for **vector**.

Linear operations

Both 2D and 3D points maintain linear space, which means that for them sum of points and multiplication of point by some number are defined. Here are those basic implementations for 2D:

```
struct point2d {
    ftype x, y;
    point2d() {}
    point2d(ftype x, ftype y): x(x), y(y) {}
    point2d& operator+=(const point2d &t) {
        x += t.x;
        y += t.y;
        return *this;
    }
    point2d& operator-=(const point2d &t) {
        x -= t.x;
        y -= t.y;
        return *this;
    }
    point2d& operator*=(ftype t) {
        x *= t;
        y *= t;
        return *this;
    }
    point2d& operator/=(ftype t) {
        x /= t;
        y /= t;
        return *this;
    }
    point2d operator+(const point2d &t) const {
        return point2d(*this) += t;
    }
    point2d operator-(const point2d &t) const {
        return point2d(*this) -= t;
    }
}
```